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Chapter 2

Hardware & Software

(1) Summary

1. Ingot Growth & wafer slicing: Pure silicon is melted into crystal ingot, then sliced into thin wafers and polished.
2. Photolithography & transistor fabrication: light patterns, etching & doping are used to creat
3. metal interconnect layers: Thin metal layers connect transistors to form circuit & memory
4. Dicing, Testing & Binning: the wafer is cut into individual chips, tested & Stored based on performance
5. Packaging: ~~Transisto~~ chips are mounted on protective package with a heat spreader to manage heat

key Takeaways

- Transistors are located on the Silicon die
- Aligment masks help machines position each layer accurately

(2) There are 5 major reasons:

1. AI Boom: AI companies need a huge amount of RAM for powerful servers & GPUs
2. HBM production: manufacturers are focusing on High Bandwidth memory (HBM), which is used in AI data center

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③ limited production: companies have limited manufacturing capacity, so they cannot immediately produce more RAM

④ Supply & Demand: AI companies buy memory in high quantities, leaving less supply for normal computers & laptops

⑤ manufacturing Costs: Building new semiconductor factories requires billions of Dollars and takes years

Main reason

Manufactures are prioritising high profit AI memory, reducing the supply of conventional RAM

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